



PES UNIVERSITY, BANGALORE
Department of Computer Science and Engineering

Project Synopsis / Proposal

AttendEase – Student Attendance Management System

1. Project Title and Group

Project Title: AttendEase – Student Attendance Management System

Group: Team 3 | **Course:** UE24CS341A — Software Engineering | PES University

2. Team Members and SRNs

Sl.	USN	SRN / PRN	Student Name
1	PES1UG24CS083	PES1202402135	Arundhati K
2	PES1UG24CS070	PES1202400437	Anshdeep Singh Sachdeva
3	PES1UG24CS093	PES1202400342	Atharva Ashish Vyas
4	PES1UG25CS807	PES1202502970	Bhoomika Dayanand Jituri

3. Problem Statement

Attendance tracking in most academic departments is still a largely manual process — faculty maintain paper registers or disconnected spreadsheets, students have no reliable way to check their own attendance standing, and generating shortage/defaulters reports at the end of a semester is slow and error-prone. There is no single system that lets administrators configure courses and sections, faculty record attendance per session, students self-monitor their attendance, and discrepancies get corrected through an auditable workflow.

4. Proposed Project Description

AttendEase is a web-based Student Attendance Management System that digitizes the full attendance lifecycle for a department. Administrators configure courses, sections, timetables, and faculty assignments. Faculty start attendance sessions and mark students present or absent for each scheduled period. Students view their own attendance dashboard in real time and can raise correction requests against disputed records, which faculty review and approve or reject. The system also computes attendance-shortage thresholds and generates defaulter and summary reports for faculty and administrators.

5. Target Users and Stakeholders

- Students — view personal attendance and raise correction requests.
- Faculty — mark attendance, review correction requests, generate reports for their sections.

- Administrators — configure courses, sections, timetables, and faculty assignments.

6. Objectives

- Replace manual/paper-based attendance tracking with a reliable, auditable digital system.
- Give students real-time visibility into their own attendance standing.
- Provide faculty a fast, low-friction way to mark attendance for a full section.
- Support a transparent, auditable correction-request workflow between students and faculty.
- Automate attendance-shortage identification and reporting for administrators.

7. Scope

In Scope	Out of Scope
Course, section, and timetable configuration	Fee management / financial systems
Session-based attendance marking by faculty	Biometric or RFID-based attendance capture
Student attendance dashboard and history	Mobile native apps (web-only for this scope)
Correction request submission and approval workflow	Multi-department / multi-institution deployment
Attendance-shortage identification and defaulter reports	Learning management / gradebook functionality

8. Functional Features and Team-wise Ownership

Each of the four team members owns one functional role end-to-end (requirements through testing) and takes on a secondary cross-cutting responsibility.

Member	USN	Role	Owned Functionality	Secondary Responsibility
Arundhati K	PES1UG24CS083	Academic & Attendance Setup	Student & academic management + Attendance recording	Database design & data validation
Anshdeep Singh Sachdeva	PES1UG24CS070	Attendance Updates & Corrections	Attendance management + Attendance correction requests	API integration & code review
Atharva Ashish Vyas	PES1UG24CS093	User Access & Student Portal	Authentication & role-based access + Student attendance dashboard	UI/UX & frontend support
Bhoomika Dayanand Jituri	PES1UG25CS807	Attendance Analytics & Reporting	Attendance shortage identification + Attendance reports	Testing, CI/CD, Jira & documentation

9. Plan of Work

Sprint	Duration	Planned Work
Sprint 1	Week 1–2	Requirements gathering, Synopsis, and SRS drafting; repository and CI/CD setup.
Sprint 2	Week 3–4	Database schema design; Auth module and Academic Setup module

Sprint	Duration	Planned Work
		scaffolding.
Sprint 3	Week 5–6	Attendance recording module; Student dashboard; initial API integration.
Sprint 4	Week 7–8	Correction-request workflow; SAD and Test Plan documentation.
Sprint 5	Week 9–10	Analytics/reporting module; security validation; integration testing.
Sprint 6	Week 11–12	System testing, bug fixes, final documentation, and demo preparation.

10. Proposed Technology and Tools

- Backend: Python 3.11, Django 4.x, Django REST Framework
- Database: PostgreSQL 14+
- Frontend: Django Templates, Bootstrap 5
- Version Control & CI/CD: Git/GitHub, GitHub Actions
- Containerization: Docker, docker-compose
- Project Tracking: Jira / GitHub Projects

11. Agile Justification

AttendEase is developed using an Agile methodology with six sprints over the semester. Each sprint delivers a working increment, reviewed and adjusted based on feedback before the next sprint begins. This fits the project well: requirements are reasonably well understood up front but implementation details (e.g., exact UI flows, report formats) benefit from iterative refinement; the four-person team can work on largely independent, owned modules in parallel; and weekly Jira/GitHub Projects tracking gives visibility into individual contribution, which is required for the course's per-member evaluation.